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 Studierichting: Informatica
 Oefeningenreeks 6G nr. 2.8

$$f(x) = e^{|x|} \quad \text{op } [-\pi, \pi]$$

$$f(-x) = e^{|-x|} = e^{|x|} = f(x) \implies \text{even}$$

$$a_n = \frac{2}{\pi} \int_0^\pi e^x \cos(nx) \cdot dx$$

$$a_0 = \frac{2}{\pi} \int_0^\pi e^x \cdot dx = \frac{2}{\pi} [e^x]_0^\pi = \frac{2(e^\pi - 1)}{\pi}$$

$$I = \int e^x \cos(nx) \cdot dx$$

$$\begin{cases} u = \cos(nx) \\ dv = e^x \cdot dx \end{cases} \Rightarrow \begin{cases} du = -n \sin(nx) \cdot dx \\ v = e^x \end{cases}$$

$$I = e^x \cos(nx) + n \int e^x \sin(nx) \cdot dx$$

$$\begin{cases} u = \sin(nx) \\ dv = e^x \cdot dx \end{cases} \Rightarrow \begin{cases} du = n \cos(nx) \cdot dx \\ v = e^x \end{cases}$$

$$I = e^x \cos(nx) + n \left(e^x \sin(nx) - n \int e^x \cos(nx) \cdot dx \right)$$

$$I = e^x \cos(nx) + ne^x \sin(nx) - n^2 I$$

$$(1 + n^2) I = e^x (\cos(nx) + n \sin(nx))$$

$$I = \frac{e^x}{1 + n^2} (\cos(nx) + n \sin(nx))$$

$$a_n = \frac{2}{\pi} \left[\frac{e^x}{1 + n^2} (\cos(nx) + n \sin(nx)) \right]_0^\pi$$

$$a_n = \frac{2}{\pi (1 + n^2)} (e^\pi (\cos(n\pi) + n \sin(n\pi)) - e^0 (\cos(0) + n \sin(0)))$$

$$a_n = \frac{2}{\pi (1 + n^2)} ((-1)^n e^\pi - 1)$$

$$f(x) = \frac{e^\pi - 1}{\pi} + \sum_{n=1}^{\infty} \frac{2((-1)^n e^\pi - 1)}{\pi (n^2 + 1)} \cos(nx)$$

References